LOGARITHM PROPERTIES

The one-to-one property

Click the hypothesis to see what it does and what breaks without it.

IF

1. $\log_a(M) = \log_a(N)$: a single logarithm on each side, same base

   CLICK: DROP IT

$\Longrightarrow$

THEN

$$M = N$$

Logarithmic functions are strictly monotonic, so they cannot assign the same output to two different inputs. $\log_3(2x + 1) = \log_3(x + 5)$ gives $2x + 1 = x + 5$, so $x = 4$.